

Power Budget A1					
Team Number:					
Project Name:					
Team Member Names:					
Version:					

A. List ALL major components (active devices, integrated circuits, etc.) except for power sources, voltage regulators, resistors, capacitors, or passive elements

All Major Components	Component Name	Part Number	Supply Voltage Range	#	Absolute Maximum Current (mA) [1]	Total Current (mA)	Unit
	ESP32	1965-ESP32-S3-WROOM-1-N4CT-ND	3-3.6 V	1	379	379	mA
	OLED Screen	I2C IIC SSD1306	3-5 V	1	30	30	mA
							mA
							mA
							mA
							mA

B. Assign each major component above to ONE power rail below. Try to minimize the number of different power rails in the design. Add additional power rails or change the power rail voltages if needed.

3.3V Power Rail	Component Name	Part Number	Supply Voltage Range	#	Absolute Maximum Current (mA)	Total Current (mA)	Unit
	ESP32	1965-ESP32-S3-WROOM-1-N4CT-ND	3-3.6 V	1	379	379	mA
	OLED Screen	I2C IIC SSD1306	3-5 V	1	30	30	mA
							mA
							mA
	Subtotal					409	mA
	Safety Margin					25%	
	Total Current Required on +3.3V Rail					511.25	mA

C. For each power rail above, select a specific voltage regulator using the same process as for major component selection. Confirm that the Total Remaining Current Available on each rail above is not negative.

D. Select a specific external power source (wall supply or battery) for your system, and confirm that it can supply all of the regulators for all of the power rails simultaneously. If you need multiple power sources, list each separately below and indicate which regulators will be connected to each supply. Confirm that the Total Remaining Current Available on each power source below is not negative.

External Power Source 1	Component Name	Part Number	Supply Voltage Range	Output Voltage	Absolute Maximum Current (mA)	Total Current (mA)	Unit
Power Source 1 Selection	Plug-in Wall Supply	(full part number)	110VAC	9V	5000	5000	mA
	Total Remaining Current Available on External Power Source 1					4488.75	mA
					Battery Life	na	hours

Notes

External Supply Voltage should be determined by the dropout voltage for highest-voltage regulator (e.g., +14V for a +12V regulator).

If you have multiple units in your design (e.g., a base unit and remote unit) then you need a separate power budget for each unitA90

[1]							
For inductive loads (e.g., motors, solenoids) this is often called "stall current" on the data sheet							